

AWS RDS 500 Q&A – Categorized Cheat sheet

1. RDS Basics & Administration (1–50)

- Instance classes, storage, IOPS tuning
- Multi-AZ vs Single-AZ
- Snapshots & PITR
- Parameter groups & option groups
- RDS Proxy, SSL enforcement
- Failover basics

2. Performance & Troubleshooting (51–100)

- Query optimization
- IOPS/Disk queue issues
- High CPU/Memory scenarios
- Connection storms, pool tuning
- Replication lag root causes
- Slow snapshot restore

3. Aurora Deep Dive (101–160)

- Aurora storage scaling (10GB increments, 128TB max)
- Writer vs Reader endpoints
- Failover priorities
- Aurora Global DB & replication lag
- Aurora Parallel Query & Serverless v2
- Aurora vs RDS MySQL/Postgres

4. PostgreSQL on RDS (161–210)

- Autovacuum tuning, pg_stat monitoring
- Logical replication, DMS migrations
- Connection limits, RDS Proxy usage
- FDW, pg_cron, pgaudit support
- Disk usage from WAL buildup
- Oracle → Postgres migration challenges

5. MySQL on RDS (211–250)

- Replication lag, GTID-based replication
- Binlog retention, schema migrations
- Temp table/disk issues
- Read replica promotion
- MySQL audit plugin support
- Migration from 5.7 → 8.0

6. SQL Server on RDS (251–280)

- RDS SQL Agent jobs
- Multi-AZ via Always On
- Restoring .bak from S3
- Extended Events for deadlocks
- TempDB management
- CDC & Enterprise-only features

7. Oracle on RDS (281–310)

- Data Guard in Multi-AZ
- RMAN usage limitations
- Option groups for AWR
- Data Pump integration with S3
- Licensing (BYOL vs License Included)
- RAC not supported in RDS

8. DevOps & Automation (311–360)

- RDS Terraform resources (aws_db_instance, aws_db_parameter_group)
- CloudFormation vs Terraform
- Secrets Manager for password rotation
- Blue/green deployments
- Schema migrations via Flyway/Liquibase
- Chaos/failover testing via APIs

9. Multi-Region DR & HA (361–400)

- RDS cross-region read replicas
- Aurora Global DB (RTO/RPO)
- DNS failover strategies
- Cost optimization in replication
- Multi-region pitfalls (write conflicts, DNS latency)

10. Compliance & Security (401–430)

- HIPAA, PCI DSS, GDPR on RDS
- KMS encryption (CMK) for snapshots
- IAM authentication
- GuardDuty & anomalous login detection
- SSL enforcement, row-level data masking
- Auditing for sensitive data

11. Monitoring, SRE & Observability (431–470)

- CloudWatch anomaly detection
- Performance Insights for slow queries
- ReplicaLag monitoring
- RDS logs → CloudWatch → ELK
- Chaos testing & error budgets
- Key metrics (DiskQueueDepth, WriteLatency)

12. Real Customer Outages & Lessons (471–500)

- Patching windows causing downtime
- Outages due to storage full, long transactions, WAL buildup
- Failover DNS propagation delays
- Snapshot restore performance issues
- SSL certificate expirations
- Human misconfigurations as root cause

Usage Tips for Interview Prep

- If asked generic RDS questions → start from 1–50.

- If asked Aurora-specific → jump to 101–160.
- If PostgreSQL DBA background → 161–210.
- If MySQL-heavy → 211–250.
- If SQL Server DBA → 251–280.
- If Oracle DBA → 281–310.
- If interviewer is cloud/devops-focused → 311–360.
- If they ask DR/HA patterns → 361–400.
- If compliance/security auditors → 401–430.
- If SRE role → 431–470.
- If scenario-based outages → 471–500.

Real-Time Scenario-Based Interview Questions & Answers on AWS RDS (Amazon Relational Database Service).

Major aspects: Basics, Instances, Backups, Replication, Failover, Monitoring, Performance, Security, Cost Optimization, Multi-AZ, Read Replicas, Aurora, PostgreSQL/MySQL/SQL Server/Oracle/MariaDB engines, Migration, Troubleshooting, DevOps/Automation, and Real-World Scenarios.

1–50: Basics & Core Concepts

1. **Q:** What is AWS RDS?
A: A managed relational database service supporting multiple engines (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora). AWS handles provisioning, patching, backups, scaling.
2. **Q:** Scenario: Your team wants to reduce DBA overhead for MySQL databases. Which AWS service do you suggest?
A: Use **Amazon RDS for MySQL** to automate patching, backups, and monitoring.
3. **Q:** Difference between RDS and EC2-hosted database?
A: RDS = managed service with automation, limited OS control. EC2 = full control, manual maintenance.
4. **Q:** Scenario: Your compliance team requires encrypted databases. Does RDS support this?
A: Yes, encryption at rest (KMS) and in transit (TLS).
5. **Q:** Which DB engines does RDS support?
A: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora.
6. **Q:** Scenario: You want a serverless relational DB that auto-scales. Which option?
A: **Aurora Serverless v2**.
7. **Q:** Does RDS allow SSH into DB host?
A: No, AWS manages the host. You only connect via database endpoint.
8. **Q:** Scenario: You need multiple environments (dev, test, prod). Can you isolate them in RDS?
A: Yes, using **VPC + subnet groups + security groups**.
9. **Q:** What is an RDS DB Instance?
A: A dedicated database environment running in AWS.
10. **Q:** Scenario: A customer asks for multiple databases in one instance. Is it possible?
A: Yes, each engine supports multiple schemas/databases, subject to engine limitations.

51–100: Storage & Backup

11. **Q:** What are RDS storage types?
A: General Purpose (gp2/gp3), Provisioned IOPS (io1/io2), Magnetic (deprecated).

12. **Q:** Scenario: You need predictable I/O for 50K TPS. Which storage?
A: **io1/io2 Provisioned IOPS.**
13. **Q:** How do backups work in RDS?
A: Automated backups (point-in-time recovery) + manual snapshots.
14. **Q:** Scenario: Your backup retention is 7 days. A table was dropped 6 days ago. Can you recover?
A: Yes, use **PITR** to restore to the time before drop.
15. **Q:** What is snapshot copy?
A: Copying RDS snapshots across regions/accounts.
16. **Q:** Scenario: DR strategy requires database in another region. How?
A: Copy snapshot to target region → restore DB.
17. **Q:** What's the max retention for automated backups?
A: 35 days.
18. **Q:** Scenario: Database size is 10 TB. Backup window is limited. What to use?
A: **Aurora continuous backups to S3** for efficiency.
19. **Q:** Can you delete manual snapshots automatically?
A: No, manual snapshots must be manually managed.
20. **Q:** Scenario: You accidentally deleted an RDS instance. Can you recover?
A: Yes, restore from last snapshot/backup.

101–150: High Availability & Multi-AZ

21. **Q:** What is Multi-AZ in RDS?
A: Synchronous replication to standby in another AZ for HA.
22. **Q:** Scenario: Your RDS must survive AZ outage. What do you enable?
A: Multi-AZ deployment.
23. **Q:** Difference between Multi-AZ vs Read Replica?
A: Multi-AZ = HA/failover. Read Replica = scale-out reads.
24. **Q:** Scenario: Application requires automatic failover. Multi-AZ or Read Replica?
A: **Multi-AZ.**
25. **Q:** How does failover happen?
A: Standby becomes primary, DNS endpoint automatically updated.
26. **Q:** Scenario: Application requires zero downtime patching. Which option?
A: Use **Multi-AZ** with failover.

27. **Q:** Can you manually failover RDS?

A: Yes, reboot with failover option.

28. **Q:** Scenario: Multi-AZ enabled, standby is upgraded. Will downtime occur?

A: Minimal downtime when primary fails over.

29. **Q:** Is Multi-AZ supported for all engines?

A: Yes, except some SQL Server editions.

30. **Q:** Scenario: You need read scalability + HA. What's the solution?

A: Multi-AZ + Read Replicas.

151–200: Read Replicas

31. **Q:** What is a Read Replica?

A: An async copy of DB for scaling reads.

32. **Q:** Scenario: Analytics workload needs separation from production. Solution?

A: Use **Read Replicas**.

33. **Q:** Can Read Replicas be promoted?

A: Yes, to standalone DB.

34. **Q:** Scenario: You need DR in another region with minimal RPO. How?

A: Cross-region read replica.

35. **Q:** What is replication lag?

A: Delay between primary writes and replica apply.

36. **Q:** Scenario: Report shows stale data in replica. Cause?

A: Replication lag due to heavy writes.

37. **Q:** Max number of read replicas per instance?

A: Up to 15 (engine-dependent).

38. **Q:** Can you chain replicas?

A: Yes, in some engines like MySQL.

39. **Q:** Scenario: Read replica needs stronger consistency. Solution?

A: Aurora replicas with **reader endpoint** (low lag).

40. **Q:** Is Multi-AZ same as Read Replica?

A: No, Multi-AZ = HA, Read Replica = scaling.

201–250: Performance Tuning

41. **Q:** How do you monitor RDS performance?
A: CloudWatch metrics, Enhanced Monitoring, Performance Insights.
42. **Q:** Scenario: Queries slowing down daily. How to troubleshoot?
A: Use **Performance Insights** → identify top SQL queries.
43. **Q:** Which parameters impact RDS performance?
A: max_connections, buffer/cache settings, IOPS.
44. **Q:** Scenario: RDS CPU 90%. What's your approach?
A: Identify query hotspots, increase instance size, optimize indexes.
45. **Q:** Can you change RDS instance size without downtime?
A: Some require reboot; Aurora supports online scaling.
46. **Q:** Scenario: Database is I/O bound. Fix?
A: Upgrade to provisioned IOPS storage.
47. **Q:** What's RDS Performance Insights?
A: A tool for query-level performance analysis.
48. **Q:** Scenario: App sees "too many connections" error. Fix?
A: Increase max_connections parameter, use connection pooling.
49. **Q:** How to scale RDS vertically?
A: Change instance class.
50. **Q:** How to scale horizontally?
A: Add Read Replicas.

251–300: Security

51. **Q:** How does RDS secure data at rest?
A: KMS encryption.
52. **Q:** Scenario: PCI compliance requires encrypted backups. How?
A: Enable RDS encryption at creation (covers snapshots & backups).
53. **Q:** Does RDS support IAM authentication?
A: Yes, for MySQL and PostgreSQL.
54. **Q:** Scenario: Password rotation needed every 30 days. How?
A: Store credentials in Secrets Manager with auto-rotation.
55. **Q:** How to restrict RDS access?
A: Security groups, subnet groups, IAM roles.

56. **Q:** Scenario: You want private DB only. How?

A: Place RDS in private subnet, no public endpoint.

57. **Q:** Can RDS be accessed over SSL/TLS?

A: Yes, SSL certificates provided.

58. **Q:** Scenario: Audit team asks for query logging. How?

A: Enable engine logs → push to CloudWatch Logs.

59. **Q:** How to enforce encryption in transit?

A: Require SSL connections via parameter groups.

60. **Q:** Scenario: Customer data must stay in one region. How?

A: Disable cross-region replication, enforce regional policies.

301–350: Maintenance & Upgrades

61. **Q:** How does patching work in RDS?

A: AWS applies patches in maintenance window.

62. **Q:** Scenario: You want zero-downtime upgrade. Solution?

A: Use Multi-AZ failover during patch.

63. **Q:** Can you skip RDS patches?

A: Yes, until forced upgrades for critical issues.

64. **Q:** Scenario: Engine upgrade needed (Postgres 12 → 14). How?

A: Create snapshot → test upgrade in staging → apply in prod.

65. **Q:** How to schedule maintenance?

A: Configure **preferred maintenance window**.

66. **Q:** Scenario: Upgrade failed. What to do?

A: Restore from snapshot, retry upgrade.

67. **Q:** Are automatic minor upgrades supported?

A: Yes, option available.

68. **Q:** Scenario: Your DB is in heavy use, how to apply patch?

A: Use Multi-AZ for minimal downtime.

69. **Q:** Can you roll back engine version?

A: No direct rollback; restore snapshot from old version.

70. **Q:** Scenario: RDS needs new parameters. How to apply?

A: Modify parameter group → reboot DB if static parameters.

351–400: Migration & Integration

71. **Q:** How to migrate on-prem DB to RDS?
A: Use **AWS DMS** or backup/restore methods.
72. **Q:** Scenario: SQL Server on-prem → RDS SQL Server migration. Tool?
A: AWS DMS + Schema Conversion Tool.
73. **Q:** Can you import/export large DBs?
A: Yes, via S3 import/export.
74. **Q:** Scenario: Oracle DBA asks for full control of Oracle features. Does RDS support all?
A: No, some features like RAC not supported; use EC2 Oracle instead.
75. **Q:** How to replicate on-prem DB to RDS?
A: DMS with CDC enabled.
76. **Q:** Scenario: Analytics requires near real-time sync. How?
A: Use DMS ongoing replication.
77. **Q:** Can RDS integrate with Lambda?
A: Yes, via VPC connectivity or Data API (Aurora).
78. **Q:** Scenario: Business requires data lake sync. How?
A: Export RDS data to S3 with DMS.
79. **Q:** Can RDS connect to external apps?
A: Yes, with proper VPC, security groups, endpoints.
80. **Q:** Scenario: Migration must minimize downtime. Approach?
A: DMS + CDC replication + cutover.

401–450: Cost Optimization

81. **Q:** How to reduce RDS cost?
A: Reserved instances, stop-start, right-sizing.
82. **Q:** Scenario: DB used only in office hours. Optimize cost?
A: Stop RDS after hours.
83. **Q:** What are reserved instances?
A: 1–3 year commitments for up to 70% discount.
84. **Q:** Scenario: Startup wants flexible billing. Which pricing?
A: On-demand.
85. **Q:** Can you stop Multi-AZ instances?
A: No, only single-AZ.

86. **Q:** Scenario: Storage cost is rising. What to do?

A: Use **storage auto-scaling** with gp3.

87. **Q:** Can snapshots reduce cost?

A: Yes, you can delete DB and retain snapshots.

88. **Q:** Scenario: You need Aurora but cost concern. Option?

A: Aurora Serverless v2 scales per second.

89. **Q:** How does pricing differ between RDS engines?

A: Aurora is premium, MySQL/Postgres cheaper, Oracle/SQL Server licensed.

90. **Q:** Scenario: You must forecast DB costs. Tool?

A: AWS Cost Explorer + Trusted Advisor.

451–500: Troubleshooting & Advanced Scenarios

91. **Q:** Scenario: RDS instance stuck in “modifying.” Fix?

A: Check pending parameter changes, reboot if needed.

92. **Q:** DB is slow after scaling. Why?

A: Cold cache, warm-up required.

93. **Q:** Scenario: Application fails with timeout. Troubleshoot?

A: Check SG rules, VPC routing, engine logs.

94. **Q:** RDS disk space full. Fix?

A: Enable storage autoscaling, cleanup logs.

95. **Q:** Scenario: Replication stopped in replica. Why?

A: Network issue, high lag, configuration error.

96. **Q:** DB fails to start. What’s next?

A: Restore snapshot, open support case.

97. **Q:** Scenario: CPU 100% during peak load. Fix?

A: Scale up instance, tune queries.

98. **Q:** Can you force failover in Aurora?

A: Yes, via cluster failover.

99. **Q:** Scenario: Need cross-account RDS access. How?

A: Share snapshot + restore in target account.

100. **Q:** DB parameter changes not applied. Why?

A: Some require reboot.

Aurora Deep Dive (101–160)

101. Q: You migrated from MySQL RDS to Aurora MySQL and queries are slower. How do you debug?

A: Check parameter differences (aurora_read_io_threads, innodb_buffer_pool_instances), slow query log, replication lag between writer and reader, and Aurora's Performance Insights.

102. Q: How does Aurora storage auto-scaling reduce admin overhead compared to MySQL RDS?

A: Aurora storage automatically grows in 10GB increments up to 128TB; RDS MySQL requires manual EBS resizing.

103. Q: During failover in Aurora, your app saw write conflicts. Why?

A: Application was writing to the **reader endpoint** instead of the **writer endpoint**. Fix by using writer endpoint for all write transactions.

104. Q: In Aurora global database, replication lag is ~30 seconds. Why?

A: Cross-region async replication depends on network latency, size of redo logs, and commit rates.

105. Q: Aurora MySQL backups run slowly. How is backup different in Aurora?

A: Aurora continuously backs up to S3 at the storage layer. Slow queries during backup may indicate high write I/O rather than backup overhead.

106. Q: How do you scale Aurora for high read workloads?

A: Add Aurora read replicas, use **Aurora Reader endpoint**, and enable **Query Cache** or Aurora Parallel Query.

107. Q: Aurora cluster scaling has failed due to "insufficient capacity." Why?

A: Not enough instance types available in the AZ. Fix by using different instance family or multi-AZ deployment.

108. Q: Aurora Postgres instance crashed after parameter change. How do you revert?

A: Revert parameters via parameter group rollback or clone snapshot to new cluster.

109. Q: Can Aurora Postgres support logical replication?

A: Yes, via rds.logical_replication parameter. Used for CDC or migration to Kafka/Redshift.

110. Q: You have Aurora Global DB with 3 regions. Failover took 3 minutes instead of expected <1. Why?

A: Data size, redo log backlog, or network bottleneck slowed recovery.

111. Q: Aurora PostgreSQL VACUUM didn't free space. Why?

A: Aurora uses storage layer block reuse; VACUUM doesn't reclaim space to OS but optimizes table visibility.

112. Q: How do you ensure Aurora replicas don't lag too much?

A: Monitor AuroraReplicaLag, tune workload, reduce long transactions, and increase instance size.

113. Q: Can Aurora MySQL cluster auto-scale compute?

A: Yes, via **Aurora Serverless v2**, which scales seamlessly based on load.

114. Q: What's Aurora "parallel query"?

A: Pushes certain queries (e.g., large scans) to Aurora storage nodes for distributed execution.

115. Q: You noticed Aurora writes are faster than RDS MySQL. Why?

A: Aurora log-structured storage avoids double-write buffer overhead.

116. Q: Can Aurora cluster span multiple AWS accounts?

A: No, but snapshots can be shared across accounts and restored.

117. Q: Aurora snapshot restore took hours. Why?

A: It's a **lazy restore**—data pages are loaded on demand, but if workload touches many blocks, it feels slow.

118. Q: What's "Aurora failover priority"?

A: Defines which replica is promoted first during failover.

119. Q: Can Aurora automatically switch writer back after failover?

A: No, failover stays until manually reverted.

120. Q: How is Aurora global DB used for DR?

A: Async replication to secondary region; failover promotes that region if primary fails.

PostgreSQL RDS (161–210)

161. Q: PostgreSQL RDS autovacuum isn't running. What do you check?

A: Parameter group (autovacuum), max_worker settings, disk pressure, and pg_stat_activity.

162. Q: Customer needs partitioning. How to handle?

A: Use native Postgres partitioned tables (v10+). Ensure supported in RDS engine version.

163. Q: Logical replication is lagging. What's the fix?

A: Increase wal_sender_timeout, tune max_wal_size, and use larger instances.

164. Q: Migration tool choice for RDS Postgres from on-prem?

A: Use DMS for schema + ongoing replication.

165. Q: How do you back up custom Postgres extensions?

A: Use pg_dump along with automated RDS backups, since not all extensions are snapshot-compatible.

166. Q: RDS Postgres connection limits reached. What's the cause?

A: Each instance size has max_connections limit. Solution: use connection pooling (RDS Proxy, PgBouncer).

167. Q: Deadlocks observed. How to troubleshoot?

A: Enable log_lock_waits, capture deadlock logs, and analyze queries.

168. Q: Why is Aurora Postgres preferred for analytics?

A: Faster replication, parallel query, and unlimited storage scaling.

169. Q: How to encrypt Postgres RDS logs at rest?

A: Logs stored in CloudWatch/S3 are encrypted by KMS.

170. Q: Can you enable Postgres FDW (Foreign Data Wrapper)?

A: Yes, if supported extension in that RDS version.

171. Q: A customer wants custom Postgres config file edits. How?

A: Only via parameter groups (cannot edit postgresql.conf directly).

172. Q: pg_repack not available on RDS. Workaround?

A: Use Aurora PostgreSQL, which has online VACUUM FULL alternatives, or dump/restore.

173. Q: How do you reduce replication lag for RDS Postgres read replica?

A: Increase IOPS, ensure queries on replica don't overload it.

174. Q: How do you monitor Postgres autovacuum performance?

A: CloudWatch metrics + pg_stat_user_tables.

175. Q: How to enforce SSL for RDS Postgres?

A: Modify parameter group rds.force_ssl=1.

176. Q: Can pg_cron run in RDS?

A: Yes, for supported versions with pg_cron extension.

177. Q: How to audit Postgres queries in RDS?

A: Use pgaudit extension.

178. Q: Migration from Oracle to RDS Postgres—what's biggest challenge?

A: Data type mismatches (NUMBER, CLOB) and PL/SQL translation.

179. Q: RDS Postgres disk full due to WALs. Why?

A: Replication stopped or backup disabled. Fix by resuming streaming/backup.

180. Q: RDS Postgres restart takes long. Why?

A: Large WAL replay or vacuuming at startup.

MySQL RDS (211–250)

211. Q: RDS MySQL replication lag is high. Why?

A: Long transactions, slow disk, or non-transactional tables.

212. Q: You need GTID-based replication. How?

A: Enable gtid_mode=ON in parameter group.

213. Q: RDS MySQL binlog retention?

A: Controlled via binlog_retention_hours.

214. Q: Why is Aurora MySQL preferred over RDS MySQL?

A: Better performance, replication, storage scaling, fault-tolerance.

215. Q: Migration from MySQL 5.7 to 8.0 RDS—how?

A: Use snapshot upgrade or dump/restore.

216. Q: Why MySQL temp tables fill up RDS disk?

A: Queries needing disk-based temp tables. Solution: tune tmp_table_size.

217. Q: Read replica cannot be promoted. Why?

A: Replica in multi-AZ mode cannot be promoted.

218. Q: RDS MySQL audit logging?

A: Enable MySQL Enterprise Audit plugin if supported.

219. Q: How to prevent accidental DROP?

A: Use IAM policy + DB grants.

220. Q: Why MySQL CPU spikes at 100%?

A: Bad queries, missing indexes.

SQL Server RDS (251–280)

251. Q: Can you run SQL Agent jobs?

A: Yes, via RDS SQL Agent feature.

252. Q: Linked servers supported?

A: No, except limited to same RDS environment.

253. Q: How do you implement HA in RDS SQL Server?

A: Multi-AZ deployment with Always On behind the scenes.

254. Q: Can you restore native .bak file?

A: Yes, via S3 integration for backup/restore.

255. Q: How do you capture deadlocks?

A: Use Extended Events in RDS.

256. Q: TempDB size issue. Solution?

A: Resize via parameter group.

257. Q: RDS SQL Server mirroring supported?

A: No, only Multi-AZ.

258. Q: Maintenance jobs for index rebuild?

A: Use SQL Agent or S3 scripts.

259. Q: Can you use CDC in RDS SQL Server?

A: Yes, but requires Enterprise edition.

260. Q: Can you use SSRS?

A: No, not supported in RDS.

Oracle RDS (281–310)

281. Q: Can you access ASM storage?

A: No, RDS abstracts storage.

282. Q: How to enable Data Guard?

A: Multi-AZ uses Oracle Data Guard automatically.

283. Q: Can you use RMAN directly?

A: Limited. Use RDS APIs for backup/restore.

284. Q: How do you apply PSU patch?

A: Only via RDS-managed patching.

285. Q: Oracle RAC in RDS?

A: Not supported.

286. Q: How to monitor AWR reports?

A: Enable option group with AWR views.

287. Q: Can you use Oracle Wallet in RDS?

A: Yes, via DB parameter groups.

288. Q: Export large schema?

A: Use Data Pump with S3 integration.

289. Q: RDS Oracle licensing modes?

A: License Included or BYOL.

290. Q: How to enable flashback?

A: Not supported in RDS.

DevOps Automation (311–360)

311. Q: How to create RDS instance via Terraform?

A: Use `aws_db_instance` resource.

312. Q: How to manage parameter groups in Terraform?

A: Use `aws_db_parameter_group`.

313. Q: CloudFormation vs Terraform for RDS?

A: Terraform better for multi-cloud; CFN tighter AWS integration.

314. Q: How to rotate DB passwords in CI/CD?

A: Use Secrets Manager auto-rotation with Lambda.

315. Q: RDS blue/green deployment?

A: Deploy duplicate instance, sync data, cutover DNS.

316. Q: How to apply schema migration?

A: Use Flyway or Liquibase in pipeline.

317. Q: Can you create snapshots with Terraform?

A: Yes, via `aws_db_snapshot`.

318. Q: How to integrate CloudWatch alarms with PagerDuty?

A: Use SNS + PagerDuty integration.

319. Q: How to automate Aurora failover testing?

A: Trigger `FailoverDBCluster` API in pipeline.

320. Q: How do you enforce compliance in Terraform for RDS encryption?

A: Use policies in Terraform Sentinel/OPA.

Multi-Region DR (361–400)

361. Q: How do you design multi-region RDS failover?

A: Use cross-region read replica + DNS failover.

362. Q: Aurora Global DB DR RTO/RPO?

A: RPO <1s, RTO <1min.

363. Q: RDS MySQL multi-region?

A: Only via cross-region replica.

364. Q: How do you test cross-region replica promotion?

A: Promote replica, update DNS, validate app.

365. Q: Can you replicate RDS snapshots cross-region?

A: Yes, manual or automatic copy.

366. Q: How to reduce cross-region replication costs?

A: Use compression, fewer replicas.

367. Q: What's biggest pitfall in multi-region RDS?

A: DNS propagation delays.

368. Q: Can RDS Oracle support multi-region standby?

A: Only via manual Data Pump export/import.

369. Q: Aurora vs RDS for global?

A: Aurora Global DB purpose-built; RDS relies on read replicas.

370. Q: How to handle write conflicts in multi-region DB?

A: Usually not supported—design app for single-writer.

Compliance & Security (401–430)

401. Q: How do you enforce HIPAA compliance in RDS?

A: Enable encryption, auditing, TLS, and sign BAA with AWS.

402. Q: PCI DSS with RDS?

A: Use network isolation, encryption, and monitoring.

403. Q: How do you enforce CMK encryption for RDS snapshots?

A: Use KMS CMK key in RDS settings.

404. Q: Can IAM auth replace DB users?

A: Yes, IAM DB authentication.

405. Q: How do you detect rogue queries?

A: Enable Enhanced Monitoring + Performance Insights.

406. Q: How to restrict RDS access to specific VPC CIDR?

A: Use security groups.

407. Q: Can you integrate RDS with GuardDuty?

A: Yes, for anomalous login detection.

408. Q: How do you enable SSL-only connections?

A: Modify parameter group and enforce in app.

409. Q: How to mask sensitive data in RDS?

A: Use DB-native features (e.g., SQL Server DDM, Postgres pgcrypto).

410. Q: How to monitor GDPR “right to be forgotten”?

A: Implement row-level deletes + audit logs.

Monitoring & SRE (431–470)

431. Q: Best way to detect RDS CPU anomalies?

A: CloudWatch anomaly detection alarms.

432. Q: How to trace slow queries?

A: Enable Performance Insights.

433. Q: Replica lag alerting?

A: CloudWatch alarm on ReplicaLag.

434. Q: How do you integrate RDS logs with ELK?

A: Export to CloudWatch Logs → ship to Elasticsearch.

435. Q: What's SRE "error budget" for RDS?

A: Defined by SLA (e.g., 99.95%), budget tracks downtime.

436. Q: How do you do chaos testing on RDS?

A: Trigger failover or simulate network isolation.

437. Q: How do you monitor query plans?

A: Use Postgres pg_stat_statements or SQL Server Extended Events.

438. Q: How to test failover without downtime?

A: Use replicas in staging and run simulated workloads.

439. Q: Which metrics indicate storage saturation?

A: DiskQueueDepth, WriteLatency.

440. Q: How to monitor IAM token failures?

A: Check CloudTrail for rds:Connect.

Customer Outages & Learnings (471–500)

471. Q: Customer outage due to RDS reboot during patch window. How avoid?

A: Use custom maintenance window aligned with low traffic.

472. Q: Customer ran out of storage. How to prevent?

A: Enable storage autoscaling.

473. Q: Latency spike after failover. Why?

A: DNS propagation and cold cache.

474. Q: Outage caused by long-running transaction blocking vacuum. Fix?

A: Break long transactions, tune idle_in_transaction_timeout.

475. Q: Replica promotion lost 5 minutes of data. Why?

A: Async replication → data loss possible.

476. Q: Customer enabled minor version upgrade, broke queries. Lesson?

A: Always test in staging first.

477. Q: Outage during snapshot restore. Why?

A: Lazy load caused I/O pressure.

478. Q: Outage from IAM auth failure. Fix?

A: Rotate IAM policies properly.

479. Q: Disk corruption in RDS?

A: AWS auto-rebuilds storage blocks in Aurora.

480. Q: How do you simulate full AZ outage?

A: Use Multi-AZ deployment.

481. Q: Customer overloaded single RDS with mixed workloads. Fix?

A: Split OLTP and reporting to separate clusters.

482. Q: CPU 100% from connection storm. Fix?

A: Add RDS Proxy.

483. Q: DNS failover took too long. Solution?

A: Reduce TTL to 30s.

484. Q: Customer deleted production RDS. Recovery?

A: Use automated backups + PITR.

****485.**

**** Q:** RDS outage from expired SSL cert. Fix?

A: Rotate RDS SSL root certs in advance.

486. Q: Replica never caught up after schema change. Why?

A: Schema not compatible, blocked replication thread.

487. Q: User accidentally dropped table. Fix?

A: PITR restore to point-in-time.

488. Q: Failover happened but app didn't reconnect. Why?

A: App not retrying with DNS writer endpoint.

489. Q: Long failover due to open transactions. Fix?

A: Commit or kill idle sessions.

490. Q: Instance stuck "modifying." Why?

A: Pending patch + backup running.

491. Q: RDS outage from hitting connection limit. Solution?

A: RDS Proxy, connection pool.

492. Q: Large queries killed I/O. Solution?

A: Offload to replica or Redshift.

493. Q: Outage during DDL. Why?

A: DDL locks entire table.

494. Q: Cost explosion in cross-region replica. Why?

A: High network egress charges.

495. Q: CloudFormation rollback left RDS stuck. Fix?

A: Manually delete orphaned RDS resources.

496. Q: Customer outage from patching SQL Server RDS. Fix?

A: Schedule patching window, test upgrades.

497. Q: App couldn't connect after upgrade. Why?

A: SSL mode mismatch, or driver incompatibility.

498. Q: Outage due to reaching max IOPS. Fix?

A: Enable provisioned IOPS, or migrate to Aurora.

499. Q: RDS instance unavailable after AZ outage. Fix?

A: Multi-AZ failover design.

500. Q: Final lesson: What's the #1 cause of RDS outages?

A: **Misconfiguration by customers** (parameter tuning, connections, patching). AWS infra rarely fails; human error dominates.